Recitation Assignment #7

November 6, 2024

You may complete this in class. However, if you are unable to do so, it is expected that you complete this for recitation next time

If you have any questions, please ask.

Today we're going to do a problem that we can't really do with pen and paper – the three body problem. You've seen a restricted version of this in one of my sample codes from a few weeks ago, and you may want to sneak a peak at how that worked.

We're going to have 2 "stars" orbiting one another. We're then going to fire a third star into the pair (with mass=1) and see how everything scatters.

In the meanwhile, we're going to look at the momentum and energy of the system.

New Concepts:

Physical:

- Potential Energy
- Kinetic Energy

Numerical:

- Dimensional Analysis

Computational:

- Plotting Data

1. First and foremost, it has been very awkward when we're doing calculations worrying about solar masses, the gravity constant, G, conversions from years to seconds, and so forth. One of the most useful things that we can do is write down dimensionless equations:

$$\vec{F}_{12} = -\frac{m_1 m_2}{{r_{12}}^2}\hat{r}_{12}$$

$$U_{12} = -\frac{m_1 m_2}{r_{12}}$$

The way this works is as follows: If all of your masses are in solar masses, all of your distances in Astronomical units, all of your velocities in units of units of earth's velocity. The "time" is the only weird unit (it ends up being in units of yr/2π). Don't worry too much about it, though. Otherwise, everything works perfectly!

So to begin with, I want you to take a look at recitation #3 that you submitted to Kendall. You can use it to remind you of the physics. This is going to be the jumping-off point for us this week, but in the end we'll have simulated this problem in a new unique way.

Now start your new program. You'll want to have two "stars" orbit one another. They should have:

- A separation of 2. The easiest way to do this is to put one of them (moving in the +y direction) at $\vec{r}_1 = \hat{\imath}$ and the other at $-\hat{\imath}$.
- A mass of 1 (each)
- You may color, give them a radius and put materials on them as you please.
- Add a trail to each.
- Set the initial velocity so that the two are in a circular orbit around one another. I shouldn't tell you this, but if you give them a speed of 0.5 (and the rightward going in the $\hat{\jmath}$ and the other going in the $-\hat{\jmath}$ direction) it'll make a nice orbit.

2. Here is where you want to look at the previous versions of this exercise. Set a reasonable timestep (I used $\Delta t = 0.01$) and a final time (I used 20), and start off a while loop:

```
while (t < tmax):
    rate(200)
    t+=dt
    #PHYSICS GOES HERE
```

What physics? Well, for each of your two stars, you'll want to compute:

$$\vec{F}_1 = -\frac{m_1 m_2}{{r_{12}}^3}(\vec{r}_1 - \vec{r}_2)$$

And similarly for the 2nd particle. And for each:

$$\vec{v} \to \vec{v} + \frac{\vec{F}}{m}\Delta t$$

$$\vec{r} \to \vec{r} + \vec{v}\Delta t$$

And update the trails as well.

3. Now it's time to check for conservation of energy. We know that energy is supposed to be conserved. To do that, we're going to make a plot. It requires you to set up a "graph" before your loop, as well as a "gcurve" to be plotted.

```
g1=graph(xmin=0,xmax=tmax)
kline=gcurve(graph=g1,color=color.red)
uline=gcurve(graph=g1,color=color.green)
totline=gcurve(graph=g1,color=color.black)
```

You'll need, of course, to set up `tmax` before this point.

At some point under your loop, you're going to have to compute the total kinetic energy for the stars. You may (of course) use the non-relativistic limit. I'm choosing to call mine "K". For star 1, it might look like:

```
K1=0.5*star1.m*pow(mag(star1.v),2)
```

And similarly for star 2. Then, just add them together.

To add a point to the plot, you may then add:

```
kline.plot(pos=(t,K))
```

4. Now do a similar calculation for the potential and total energy. They should be plotted with the same "graph" but with different "gcurve"s.

   Hint: If you add up all the (pairwise) calculations of:

$$U_{12} = -\frac{m_1 m_2}{r_{12}}$$

   You get the total potential energy.

5. Now add a third star. Make it:

   - The same size as the other two
   - Yellow
   - Make a yellow trail behind it as well.
   - m=1
   - With $\vec{r}_0 = -10\,\hat{\imath}$
   - With $\vec{v}_0 = 0.5\,\hat{\imath}$

   You'll need to compute the forces on/from the new star, along with its kinetic energy and the new potential energy. Since there are now 3 stars, you'll need to compute 3 pairs (as opposed to 1 with 2 stars) in the potential energy.
   Make sure the simulation runs for 20 units of time.
   Looks pretty weird, no?
   What happens to the energy?

6. Now play around with your code. You should adjust the timesteps to make energy as conserved as possible. How small did your timesteps get?

   Adjust the masses and initial values of the particles, until you get a particularly interesting looking scattering experiment.

   Adjust your initial values very slightly, including the mass of the incoming star, and see if you get a different result. Do you notice, for example, what typically happens with regards the larger or smaller star from the original binary?

7. As always, please have Kendall or the physics fellows come by to look at your work. Please send Kendall and Professor Cruz the link to your code when you are done.